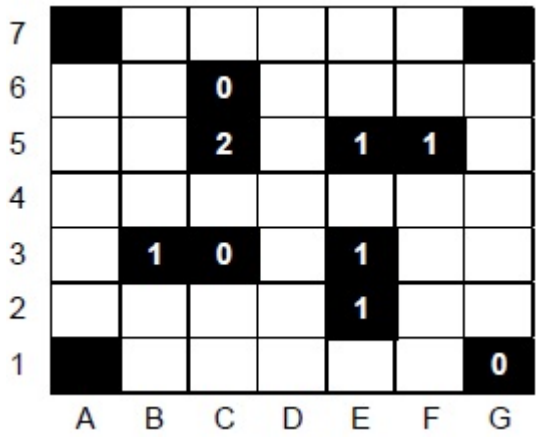
Sec 1

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

Geeta is playing Light Up which is played on a square grid. The grid has both black cells and white cells in it. The objective is to place light bulbs on the grid so that every white square is lit.
The 7 × 7 grid is as shown below:



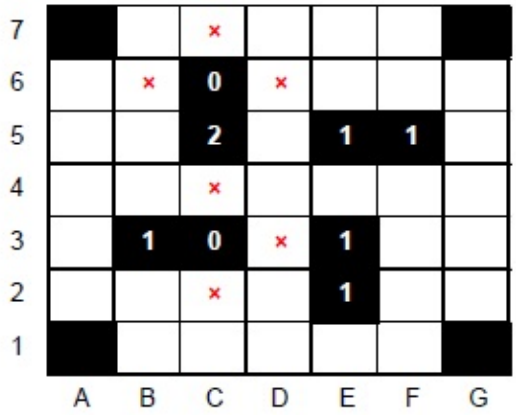
The rules of the game are:
(i) A white cell is illuminated by a light bulb placed in a cell that is in the same row or column, if there are no black cells between them.
(ii) Between any two light bulbs placed in the same row or in the same column, there has to be atleast one black cell.
(iii) Some of the black cells have numbers in them and in some the numbers have been erased. A number in a black cell indicates how many cells having the light bulbs share an edge with that cell.
(iv) The cell in which a light bulb cannot be placed is marked with 'X'.

Q.1 [11831809]
Total how many light bulbs are placed in the grid?

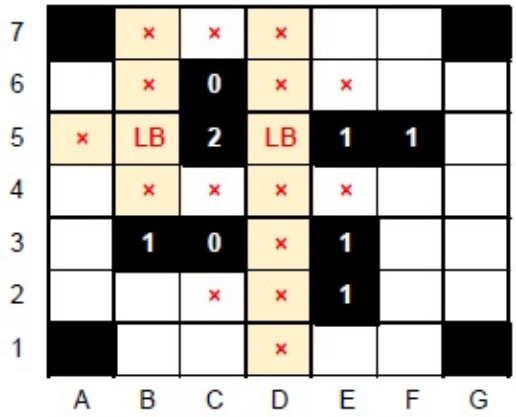
Solution:
Correct Answer : 9

Answer key/Solution

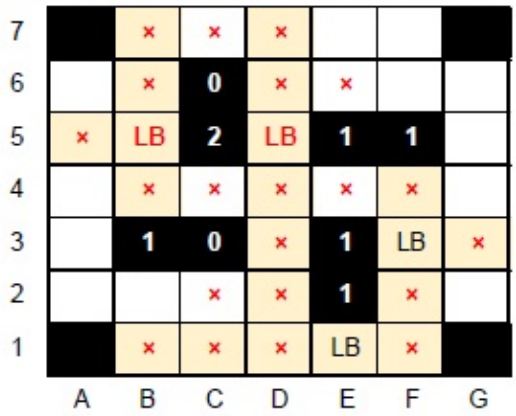
Step 1:
From condition (iii), a number in a black cell indicates how many light bulbs share an edge with that cell. Observe that C6 and C3 share 0 edges with light bulbs.



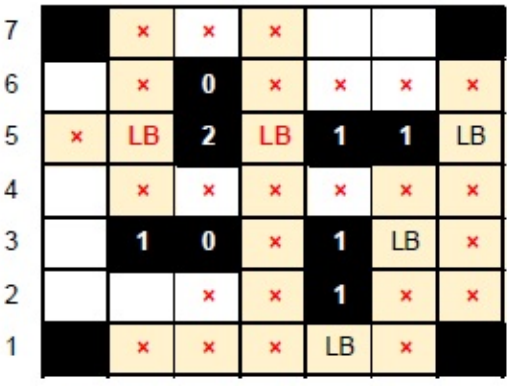
Since cell C5 share edge with two light bulbs, therefore, B5 and D5 must have light bulbs.

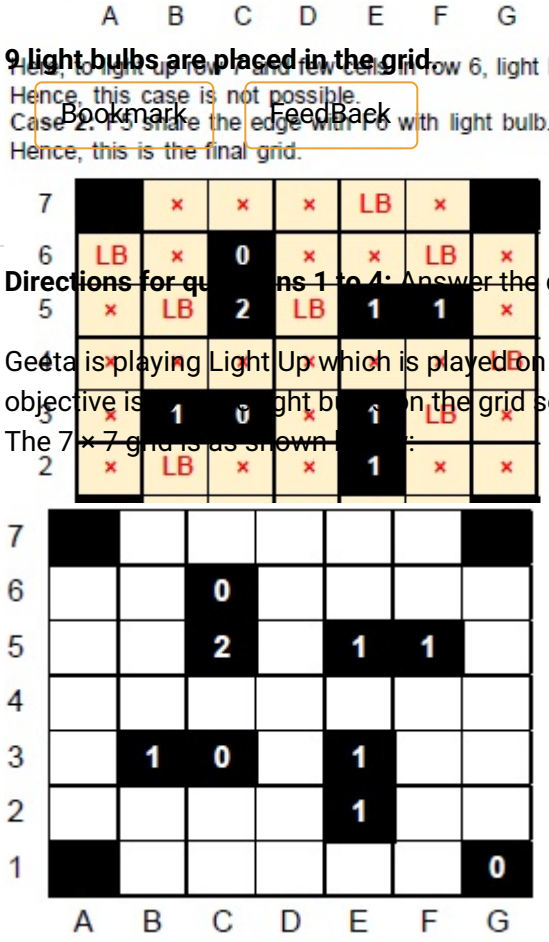


E3 share edge with one light bulb and it must be F3, which implies E2 share the edge with light bulb in cell E1.



Step 2:
Now, F5 share the edge with exactly one cell with light bulb which is either G5 or F6.
Case 1: F5 share the edge with G5 with light bulb.





9 light bulbs are placed in the grid.

Here, to light up row 7 and few cells in row 6, light bulb has to be placed in either E7 or F7 one or few cells with not be lit up.
Hence, this case is not possible.
Case 2: F5 share the edge with F6 with light bulb. This case is possible.
Hence, this is the final grid.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

Geeta is playing Light Up which is played on a square grid. The grid has both black cells and white cells in it. The objective is to light bulbs on the grid so that every white square is lit.
The 7 × 7 grid is as shown below:

The rules of the game are:
(i) A white cell is illuminated by a light bulb placed in a cell that is in the same row or column, if there are no black cells between them.
(ii) Between any two light bulbs placed in the same row or in the same column, there has to be atleast one black cell.
(iii) Some of the black cells have numbers in them and in some the numbers have been erased. A number in a black cell indicates how many cells having the light bulbs share an edge with that cell.
(iv) The cell in which a light bulb cannot be placed is marked with 'X'.

Q.2 [11831809]
Which of the following column has the minimum number of cells marked with 'X'?

1 ◯ B

2 ◯ C

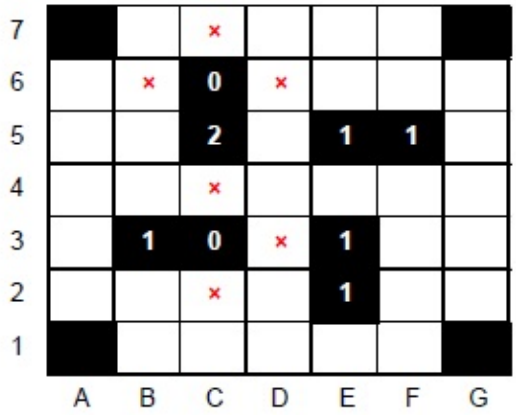
3 ◯ E

4 ◯ G

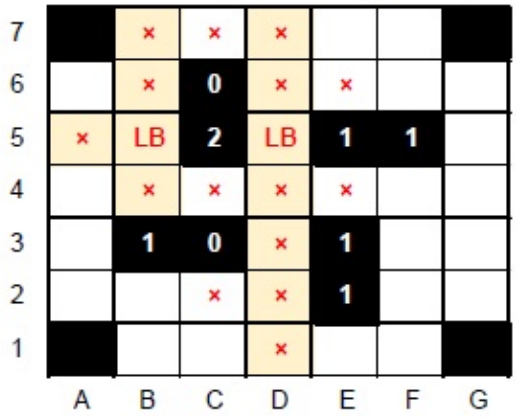
Solution:
Correct Answer : 3

Answer key/Solution

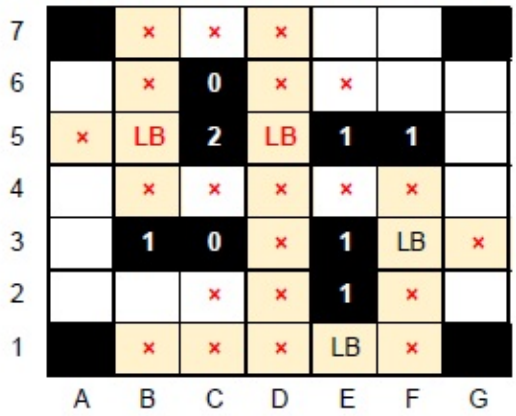
Step 1:
From condition (iii), a number in a black cell indicates how many light bulbs share an edge with that cell. Observe that C6 and C3 share 0 edges with light bulbs.



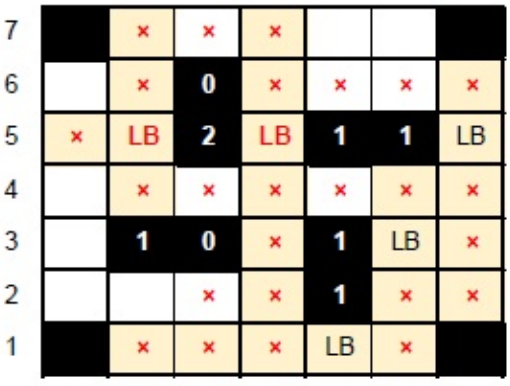
Since cell C5 share edge with two light bulbs, therefore, B5 and D5 must have light bulbs.



E3 share edge with one light bulb and it must be F3, which implies E2 share the edge with light bulb in cell E1.

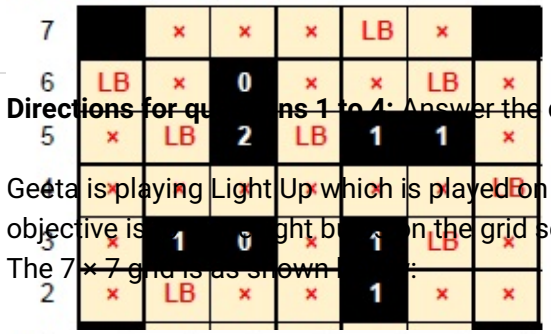


Step 2:
Now, F5 share the edge with exactly one cell with light bulb which is either G5 or F6.
Case 1: F5 share the edge with G5 with light bulb.



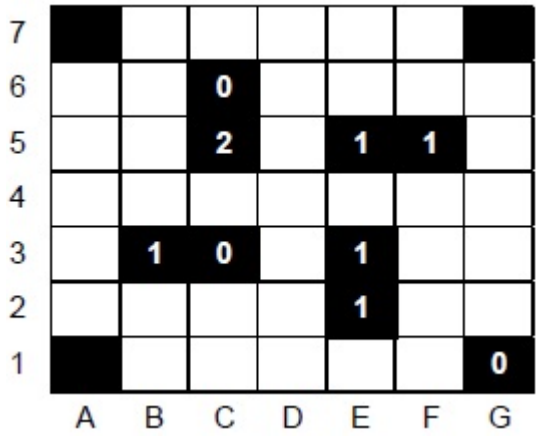
Column E has the minimum number of cells marked with 'X'.

Here, to light up row 1 and few cells in row 6, light bulb has to be placed in either E7 or F7 one or few cells with not be lit up.
Hence, this case is not possible.
Case 2: F5 share the edge with F6 with light bulb. This case is possible.
Hence, this is the final grid.



Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

Geeta is playing Light Up which is played on a square grid. The grid has both black cells and white cells in it. The objective is to place light bulbs on the grid so that every white square is lit.
The 7 × 7 grid is as shown below:



The rules of the game are:
(i) A white cell is illuminated by a light bulb placed in a cell that is in the same row or column, if there are no black cells between them.
(ii) Between any two light bulbs placed in the same row or in the same column, there has to be atleast one black cell.
(iii) Some of the black cells have numbers in them and in some the numbers have been erased. A number in a black cell indicates how many cells having the light bulbs share an edge with that cell.
(iv) The cell in which a light bulb cannot be placed is marked with 'X'.

Q.3 [11831809]

If all black cells had a number in them, then what is the sum of the numbers in all the black cells?

1 ◯ 7

2 ◯ 8

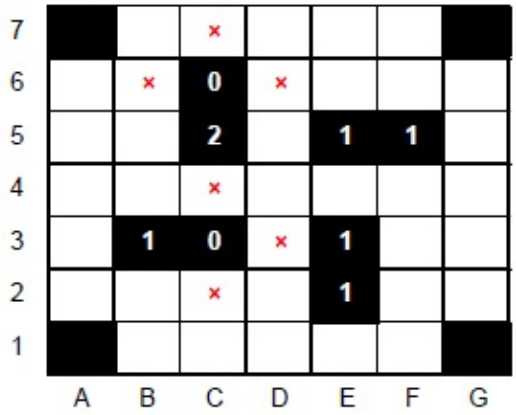
3 ◯ 9

4 ◯ Either (2) or (3)

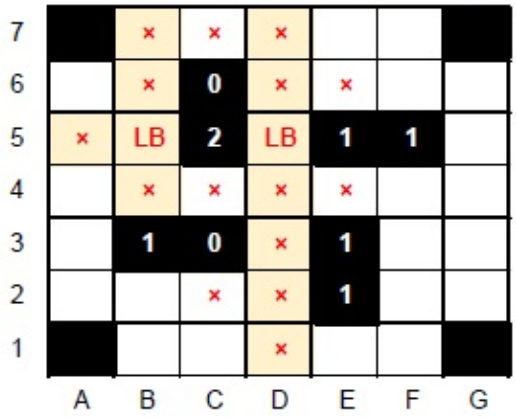
Solution:
Correct Answer : 2

Answer key/Solution

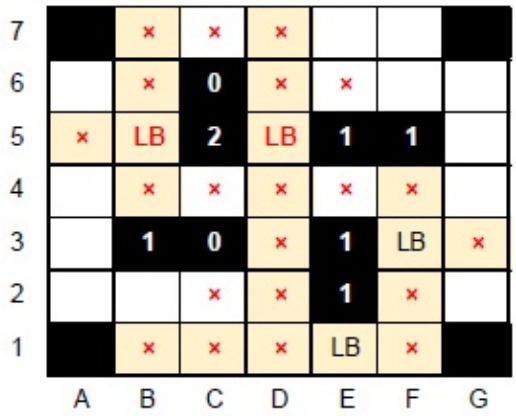
Step 1:
From condition (iii), a number in a black cell indicates how many light bulbs share an edge with that cell. Observe that C6 and C3 share 0 edges with light bulbs.



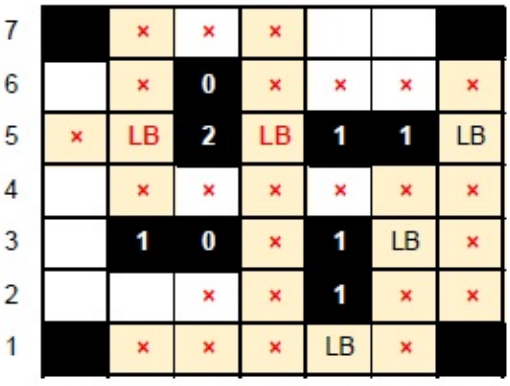
Since cell C5 share edge with two light bulbs, therefore, B5 and D5 must have light bulbs.

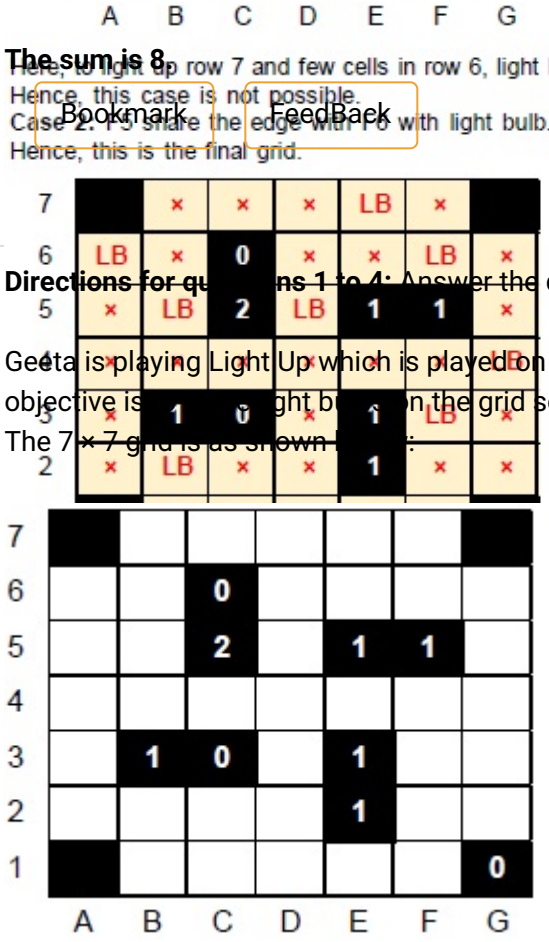


E3 share edge with one light bulb and it must be F3, which implies E2 share the edge with light bulb in cell E1.



Step 2:
Now, F5 share the edge with exactly one cell with light bulb which is either G5 or F6.
Case 1: F5 share the edge with G5 with light bulb.





The sum is 8.

Here, to light up row 7 and few cells in row 6, light bulb has to be placed in either E7 or F7 one or few cells with not be lit up.
Hence, this case is not possible.
Case 2: F5 share the edge with F6 with light bulb. This case is possible.
Hence, this is the final grid.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

Geeta is playing Light Up which is played on a square grid. The grid has both black cells and white cells in it. The objective is to light bulbs on the grid so that every white square is lit.
The 7 × 7 grid is as shown below:

The rules of the game are:
(i) A white cell is illuminated by a light bulb placed in a cell that is in the same row or column, if there are no black cells between them.
(ii) Between any two light bulbs placed in the same row or in the same column, there has to be atleast one black cell.
(iii) Some of the black cells have numbers in them and in some the numbers have been erased. A number in a black cell indicates how many cells having the light bulbs share an edge with that cell.
(iv) The cell in which a light bulb cannot be placed is marked with 'X'.

Q.4 [11831809]

Which of the following is/are true about cell F4?
I. The cell shares an edge with exactly one light bulb.
II. The cell shares an edge with exactly two light bulbs.
III. The cell is in the column with only one light bulb.

1 ○ II only

2 ○ II & III only

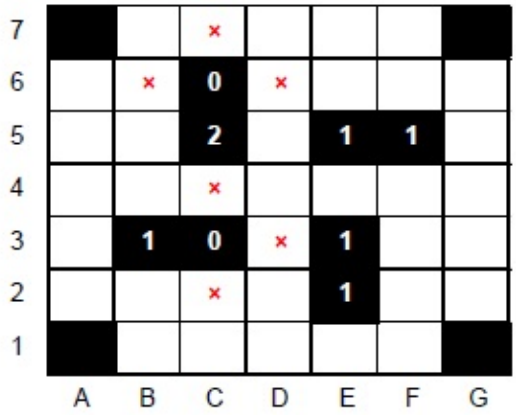
3 ○ III only

4 ○ I & III only

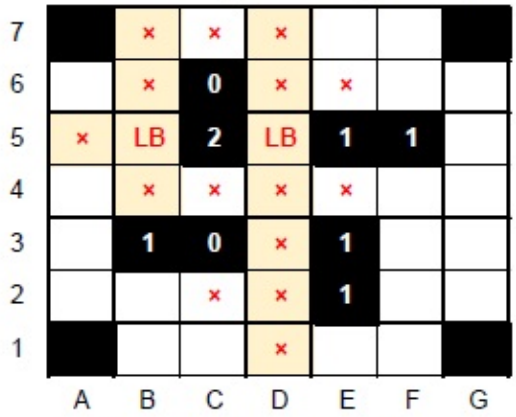
Solution:
Correct Answer : 1

Answer key/Solution

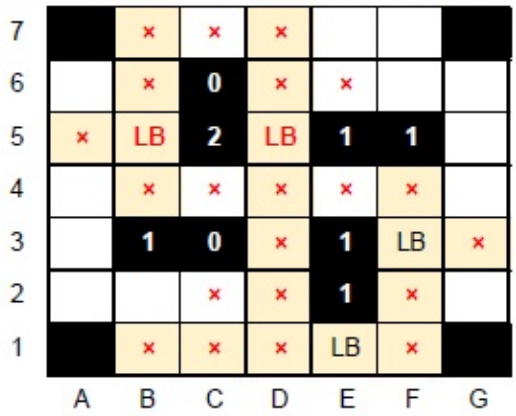
Step 1:
From condition (iii), a number in a black cell indicates how many light bulbs share an edge with that cell. Observe that C6 and C3 share 0 edges with light bulbs.



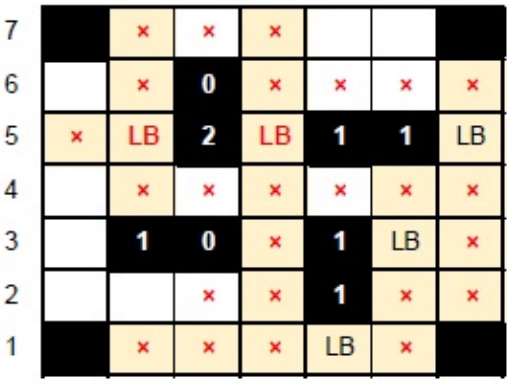
Since cell C5 share edge with two light bulbs, therefore, B5 and D5 must have light bulbs.



E3 share edge with one light bulb and it must be F3, which implies E2 share the edge with light bulb in cell E1.

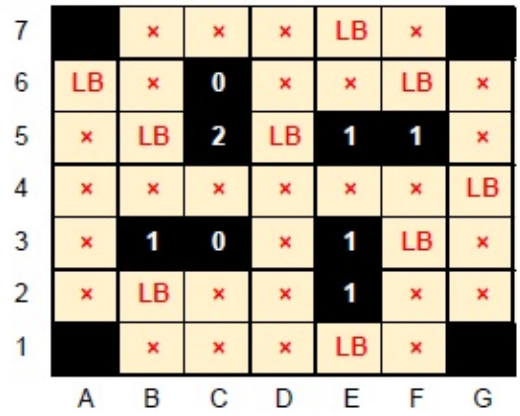


Step 2:
Now, F5 share the edge with exactly one cell with light bulb which is either G5 or F6.
Case 1: F5 share the edge with G5 with light bulb.



Cell F4 share the edge with 2 light bulbs. Hence, only statement (I) is correct.

Here, to light up row 7 and few cells in row 6, light bulb has to be placed in either E7 or F7 one or few cells with not be lit up.
Hence, this case is not possible.
Case 2: F5 share the edge with F6 with light bulb. This case is possible.
Hence, this is the final grid.



Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Jaldiram has 3 sweet outlets in Sector 31, Sector 35 and Sector 70 of Gurgaon. He specializes in 4 types of sweets - Ladoos, Barfi, Rasgulla and Rasmalai. The following table gives partial information about the sales (in integral multiples of Rs. 1,000) of each outlet for the 4 consecutive days (from October 23 to October 26) during Diwali.

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _	4, _, _, _	6, 9, _, _
October 24	4, 6, _, _	_, _, 10, 14	3, _, 9, _
October 25	6, _, _, 12	_, 7, _, 15	3, _, _, 15
October 26	_, _, 12, 15	6, 9, _, _	_, _, 11, 14
Total	144	148	144

Note: In the above table, the sales figures are in the order - Ladoos, Barfi, Rasgulla, Rasmalai.
For example: The sales of Rasgulla and Rasmalai in Sector 35 outlet on October 24 were Rs. 10,000 and Rs. 14,000 respectively.
The sales of various sweets are in the range (in Rs. '000s) as per the table below:

Type of Sweets	Range
Ladoos	From 3 to 6
Barfi	From 6 to 9
Rasgulla	From 9 to 12
Rasmalai	From 12 to 15

For example: The sales of Ladoos in any outlet on any day can be equal to or more than Rs. 3,000 but less than or equal to Rs. 6,000.
The other known information are as below:
(i) The average of the sales of all the sweets is an integer (in integral multiples of Rs. 1,000), for each of the outlets on all the 4 days.
(ii) On each day exactly two of the outlets had the same total sales turnover. Each outlet had the same total turnover on exactly two out of the 4 days.
(iii) On October 25, each of the three outlets had the same sales turnover for exactly one of the sweets.
(iv) In Sector 35 outlet, On October 23rd, 24th and 25th the sales turnover of exactly one of the sweets was the same for all the three days.
(v) The total sales of Rasmalai on all the 4 days together, was an odd number for each of the three outlets.
(vi) The sales turnover of Barfi in Sector 70 on the 4 days was in an Arithmetic Progression.
(vii) The sales of the 4 types of sweets was not all even or all odd in any of the outlets on all the 4 days. There was only one exception to this rule on a particular day at one of the outlets.

Q.5 [11831809]
The sales of Rasgulla at Sector 31 on October 24 was (in Rs. '000)

1 ◯ 13

2 ◯ 12

3 ◯ 11

4 ◯ Either (1) or (3)

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
From the given range (in Rs. '000s) of the sales of various sweets, we can see that the maximum value of total sales at any outlet on any day = 6 + 9 + 12 + 15 = 42 and minimum value = 3 + 6 + 9 + 12 = 30.
From condition (i), we can conclude that the sales turnover (in Rs. '000s) at any outlet on any day may be 32/36/40.
From condition (iii), we can conclude the sales turnover (in Rs. '000s) as per below table range of the respective sales of sweets should also be kept in mind while preparing the below table:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _ (32)	4, _, _, _ (32)	6, 9, _, _ (40)
October 24	4, 6, _, _ (36)	_, _, 10, 14 (36)	3, _, 9, _ (32)
October 25	6, _, _, 12 (36)	_, 7, _, 15 (40)	3, _, _, 15 (36)
October 26	_, _, 12, 15 (40)	6, 9, _, _ (40)	_, _, 11, 14 (36)
Total	144	148	144

Step 2:
From condition (vi), we can conclude that the sales turnover of Barfi at Sector 70 on October 23, 24, 25 and 26 must be 9, 8, 7 and 6 (in Rs. '000s) respectively.
Now, we can see that the sales of each sweet at Sector 70 on October 25 were 3, 7, 11 and 15 (in Rs. '000s) which are all odd numbers. Except these values; all other values of the sales must be 2 even and 2 odd.

So the sales at Sector 35 on October 25 must be 6, 7, 12 and 15 (in Rs. '000s).
From condition (iii), we can conclude that the sales of Barfi at each outlet must be equal i.e., 7 (in Rs. '000s).
From condition (iv), we can conclude that the sales of Barfi at Sector 35 on each of the days October 23, 24 and 25 must be 7 (in Rs. '000s).
From condition (v), the sales of Rasmalai at Sector 70 on October 23 might be 12/14 (in Rs. '000s).
But if it is 12 (in Rs. '000s), then the sales of Rasgulla will be 13 (in Rs. '000s) which is not in the range.
So, the sales of sweets at Sector 70 on October 23 must be 6, 9, 11, 14 (in Rs. '000s).

Step 3:
From condition (vii), we can conclude that the sales of sweets at Sector 31 on October 24 would be 4, 6, 11 and 15 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 31 on October 23 would be 3, 6, 10 and 13 (in Rs. '000s).
It is very clear that the sales of sweets at Sector 35 on October 23 would be 4, 7, 9 and 12 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 35 on October 26 would be 6, 9, 11 and 14 (in Rs. '000s).
Hence, the final table can be shown as:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, 10, 13 (32)	4, 7, 9, 12 (32)	6, 9, 11, 14 (40)
October 24	4, 6, 11, 15 (36)	5, 7, 10, 14 (36)	3, 8, 9, 12 (32)
October 25	6, 7, 11, 12 (36)	6, 7, 12, 15 (40)	3, 7, 11, 15 (36)
October 26	4/5/6, 9/8/7, 12, 15 (40)	6, 9, 11, 14 (40)	5, 6, 11, 14 (36)
Total	144	148	144

The sales of Rasgulla at Sector 31 on October 24 was 11 (in Rs. '000).

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Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Jaldiram has 3 sweet outlets in Sector 31, Sector 35 and Sector 70 of Gurgaon. He specializes in 4 types of sweets - Ladoos, Barfi, Rasgulla and Rasmalai. The following table gives partial information about the sales (in integral multiples of Rs. 1,000) of each outlet for the 4 consecutive days (from October 23 to October 26) during Diwali.

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _	4, _, _, _	6, 9, _, _
October 24	4, 6, _, _	_, _, 10, 14	3, _, 9, _
October 25	6, _, _, 12	_, 7, _, 15	3, _, _, 15
October 26	_, _, 12, 15	6, 9, _, _	_, _, 11, 14
Total	144	148	144

Note: In the above table, the sales figures are in the order - Ladoos, Barfi, Rasgulla, Rasmalai.
For example: The sales of Rasgulla and Rasmalai in Sector 35 outlet on October 24 were Rs. 10,000 and Rs. 14,000 respectively.
The sales of various sweets are in the range (in Rs. '000s) as per the table below:

Type of Sweets	Range
Ladoos	From 3 to 6
Barfi	From 6 to 9
Rasgulla	From 9 to 12
Rasmalai	From 12 to 15

For example: The sales of Ladoos in any outlet on any day can be equal to or more than Rs. 3,000 but less than or equal to Rs. 6,000.
The other known information are as below:
(i) The average of the sales of all the sweets is an integer (in integral multiples of Rs. 1,000), for each of the outlets on all the 4 days.
(ii) On each day exactly two of the outlets had the same total sales turnover. Each outlet had the same total turnover on exactly two out of the 4 days.
(iii) On October 25, each of the three outlets had the same sales turnover for exactly one of the sweets.
(iv) In Sector 35 outlet, On October 23rd, 24th and 25th the sales turnover of exactly one of the sweets was the same for all the three days.
(v) The total sales of Rasmalai on all the 4 days together, was an odd number for each of the three outlets.
(vi) The sales turnover of Barfi in Sector 70 on the 4 days was in an Arithmetic Progression.
(vii) The sales of the 4 types of sweets was not all even or all odd in any of the outlets on all the 4 days. There was only one exception to this rule on a particular day at one of the outlets.

Q.6 [11831809]

The total sales turnover of all outlets on October 25 was (in Rs. '000)

1 ◯ 100

2 ◯ 104

3 ◯ 112

4 ◯ 116

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
From the given range (in Rs. '000s) of the sales of various sweets, we can see that the maximum value of total sales at any outlet on any day = 6 + 9 + 12 + 15 = 42 and minimum value = 3 + 6 + 9 + 12 = 30.
From condition (i), we can conclude that the sales turnover (in Rs. '000s) at any outlet on any day may be 32/36/40.
From condition (iii), we can conclude the sales turnover (in Rs. '000s) as per below table range of the respective sales of sweets should also be kept in mind while preparing the below table:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _ (32)	4, _, _, _ (32)	6, 9, _, _ (40)
October 24	4, 6, _, _ (36)	_, _, 10, 14 (36)	3, _, 9, _ (32)
October 25	6, _, _, 12 (36)	_, 7, _, 15 (40)	3, _, _, 15 (36)
October 26	_, _, 12, 15 (40)	6, 9, _, _ (40)	_, _, 11, 14 (36)
Total	144	148	144

Step 2:
From condition (vi), we can conclude that the sales turnover of Barfi at Sector 70 on October 23, 24, 25 and 26 must be 9, 8, 7 and 6 (in Rs. '000s) respectively.
Now, we can see that the sales of each sweet at Sector 70 on October 25 were 3, 7, 11 and 15 (in Rs. '000s) which are all odd numbers. Except these values; all other values of the sales must be 2 even and 2 odd.

So the sales at Sector 35 on October 25 must be 6, 7, 12 and 15 (in Rs. '000s).
From condition (iii), we can conclude that the sales of Barfi at each outlet must be equal i.e., 7 (in Rs. '000s).
From condition (iv), we can conclude that the sales of Barfi at Sector 35 on each of the days October 23, 24 and 25 must be 7 (in Rs. '000s).
From condition (v), the sales of Rasmalai at Sector 70 on October 23 might be 12/14 (in Rs. '000s).
But if it is 12 (in Rs. '000s), then the sales of Rasgulla will be 13 (in Rs. '000s) which is not in the range.
So, the sales of sweets at Sector 70 on October 23 must be 6, 9, 11, 14 (in Rs. '000s).

Step 3:
From condition (vii), we can conclude that the sales of sweets at Sector 31 on October 24 would be 4, 6, 11 and 15 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 31 on October 23 would be 3, 6, 10 and 13 (in Rs. '000s).
It is very clear that the sales of sweets at Sector 35 on October 23 would be 4, 7, 9 and 12 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 35 on October 26 would be 6, 9, 11 and 14 (in Rs. '000s).
Hence, the final table can be shown as:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, 10, 13 (32)	4, 7, 9, 12 (32)	6, 9, 11, 14 (40)
October 24	4, 6, 11, 15 (36)	5, 7, 10, 14 (36)	3, 8, 9, 12 (32)
October 25	6, 7, 11, 12 (36)	6, 7, 12, 15 (40)	3, 7, 11, 15 (36)
October 26	4/5/6, 9/8/7, 12, 15 (40)	6, 9, 11, 14 (40)	5, 6, 11, 14 (36)
Total	144	148	144

The total sales turnover of all outlets on October 25 was 36 + 40 + 36 = 112 (in Rs. '000).

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Jaldiram has 3 sweet outlets in Sector 31, Sector 35 and Sector 70 of Gurgaon. He specializes in 4 types of sweets - Ladoos, Barfi, Rasgulla and Rasmalai. The following table gives partial information about the sales (in integral multiples of Rs. 1,000) of each outlet for the 4 consecutive days (from October 23 to October 26) during Diwali.

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _	4, _, _, _	6, 9, _, _
October 24	4, 6, _, _	_, _, 10, 14	3, _, 9, _
October 25	6, _, _, 12	_, 7, _, 15	3, _, _, 15
October 26	_, _, 12, 15	6, 9, _, _	_, _, 11, 14
Total	144	148	144

Note: In the above table, the sales figures are in the order - Ladoos, Barfi, Rasgulla, Rasmalai.
For example: The sales of Rasgulla and Rasmalai in Sector 35 outlet on October 24 were Rs. 10,000 and Rs. 14,000 respectively.
The sales of various sweets are in the range (in Rs. '000s) as per the table below:

Type of Sweets	Range
Ladoos	From 3 to 6
Barfi	From 6 to 9
Rasgulla	From 9 to 12
Rasmalai	From 12 to 15

For example: The sales of Ladoos in any outlet on any day can be equal to or more than Rs. 3,000 but less than or equal to Rs. 6,000.
The other known information are as below:
(i) The average of the sales of all the sweets is an integer (in integral multiples of Rs. 1,000), for each of the outlets on all the 4 days.
(ii) On each day exactly two of the outlets had the same total sales turnover. Each outlet had the same total turnover on exactly two out of the 4 days.
(iii) On October 25, each of the three outlets had the same sales turnover for exactly one of the sweets.
(iv) In Sector 35 outlet, On October 23rd, 24th and 25th the sales turnover of exactly one of the sweets was the same for all the three days.
(v) The total sales of Rasmalai on all the 4 days together, was an odd number for each of the three outlets.
(vi) The sales turnover of Barfi in Sector 70 on the 4 days was in an Arithmetic Progression.
(vii) The sales of the 4 types of sweets was not all even or all odd in any of the outlets on all the 4 days. There was only one exception to this rule on a particular day at one of the outlets.

Q.7 [11831809]

The sales of Barfi at Sector 31 on October 26th cannot be (in Rs. '000)

1 ◯ 6

2 ◯ 7

3 ◯ 9

4 ◯ 8

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
From the given range (in Rs. '000s) of the sales of various sweets, we can see that the maximum value of total sales at any outlet on any day = 6 + 9 + 12 + 15 = 42 and minimum value = 3 + 6 + 9 + 12 = 30.
From condition (i), we can conclude that the sales turnover (in Rs. '000s) at any outlet on any day may be 32/36/40.
From condition (iii), we can conclude the sales turnover (in Rs. '000s) as per below table range of the respective sales of sweets should also be kept in mind while preparing the below table:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _ (32)	4, _, _, _ (32)	6, 9, _, _ (40)
October 24	4, 6, _, _ (36)	_, _, 10, 14 (36)	3, _, 9, _ (32)
October 25	6, _, _, 12 (36)	_, 7, _, 15 (40)	3, _, _, 15 (36)
October 26	_, _, 12, 15 (40)	6, 9, _, _ (40)	_, _, 11, 14 (36)
Total	144	148	144

Step 2:
From condition (vi), we can conclude that the sales turnover of Barfi at Sector 70 on October 23, 24, 25 and 26 must be 9, 8, 7 and 6 (in Rs. '000s) respectively.
Now, we can see that the sales of each sweet at Sector 70 on October 25 were 3, 7, 11 and 15 (in Rs. '000s) which are all odd numbers. Except these values; all other values of the sales must be 2 even and 2 odd.

So the sales at Sector 35 on October 25 must be 6, 7, 12 and 15 (in Rs. '000s).
From condition (iii), we can conclude that the sales of Barfi at each outlet must be equal i.e., 7 (in Rs. '000s).
From condition (iv), we can conclude that the sales of Barfi at Sector 35 on each of the days October 23, 24 and 25 must be 7 (in Rs. '000s).
From condition (v), the sales of Rasmalai at Sector 70 on October 23 might be 12/14 (in Rs. '000s).
But if it is 12 (in Rs. '000s), then the sales of Rasgulla will be 13 (in Rs. '000s) which is not in the range.
So, the sales of sweets at Sector 70 on October 23 must be 6, 9, 11, 14 (in Rs. '000s).

Step 3:
From condition (vii), we can conclude that the sales of sweets at Sector 31 on October 24 would be 4, 6, 11 and 15 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 31 on October 23 would be 3, 6, 10 and 13 (in Rs. '000s).
It is very clear that the sales of sweets at Sector 35 on October 23 would be 4, 7, 9 and 12 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 35 on October 26 would be 6, 9, 11 and 14 (in Rs. '000s).
Hence, the final table can be shown as:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, 10, 13 (32)	4, 7, 9, 12 (32)	6, 9, 11, 14 (40)
October 24	4, 6, 11, 15 (36)	5, 7, 10, 14 (36)	3, 8, 9, 12 (32)
October 25	6, 7, 11, 12 (36)	6, 7, 12, 15 (40)	3, 7, 11, 15 (36)
October 26	4/5/6, 9/8/7, 12, 15 (40)	6, 9, 11, 14 (40)	5, 6, 11, 14 (36)
Total	144	148	144

The sales of Barfi at Sector 31 on October 26th cannot be 6 (in Rs. '000).

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Jaldiram has 3 sweet outlets in Sector 31, Sector 35 and Sector 70 of Gurgaon. He specializes in 4 types of sweets - Ladoos, Barfi, Rasgulla and Rasmalai. The following table gives partial information about the sales (in integral multiples of Rs. 1,000) of each outlet for the 4 consecutive days (from October 23 to October 26) during Diwali.

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _	4, _, _, _	6, 9, _, _
October 24	4, 6, _, _	_, _, 10, 14	3, _, 9, _
October 25	6, _, _, 12	_, 7, _, 15	3, _, _, 15
October 26	_, _, 12, 15	6, 9, _, _	_, _, 11, 14
Total	144	148	144

Note: In the above table, the sales figures are in the order - Ladoos, Barfi, Rasgulla, Rasmalai.
For example: The sales of Rasgulla and Rasmalai in Sector 35 outlet on October 24 were Rs. 10,000 and Rs. 14,000 respectively.
The sales of various sweets are in the range (in Rs. '000s) as per the table below:

Type of Sweets	Range
Ladoos	From 3 to 6
Barfi	From 6 to 9
Rasgulla	From 9 to 12
Rasmalai	From 12 to 15

For example: The sales of Ladoos in any outlet on any day can be equal to or more than Rs. 3,000 but less than or equal to Rs. 6,000.
The other known information are as below:
(i) The average of the sales of all the sweets is an integer (in integral multiples of Rs. 1,000), for each of the outlets on all the 4 days.
(ii) On each day exactly two of the outlets had the same total sales turnover. Each outlet had the same total turnover on exactly two out of the 4 days.
(iii) On October 25, each of the three outlets had the same sales turnover for exactly one of the sweets.
(iv) In Sector 35 outlet, On October 23rd, 24th and 25th the sales turnover of exactly one of the sweets was the same for all the three days.
(v) The total sales of Rasmalai on all the 4 days together, was an odd number for each of the three outlets.
(vi) The sales turnover of Barfi in Sector 70 on the 4 days was in an Arithmetic Progression.
(vii) The sales of the 4 types of sweets was not all even or all odd in any of the outlets on all the 4 days. There was only one exception to this rule on a particular day at one of the outlets.

Q.8 [11831809]

The total sales of Rasmalai at Sector 35 on these 4 days was equal to (in Rs. '000)

Solution:
Correct Answer : 55

Answer key/Solution

Step 1:
From the given range (in Rs. '000s) of the sales of various sweets, we can see that the maximum value of total sales at any outlet on any day = 6 + 9 + 12 + 15 = 42 and minimum value = 3 + 6 + 9 + 12 = 30.
From condition (i), we can conclude that the sales turnover (in Rs. '000s) at any outlet on any day may be 32/36/40.
From condition (iii), we can conclude the sales turnover (in Rs. '000s) as per below table range of the respective sales of sweets should also be kept in mind while preparing the below table:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _ (32)	4, _, _, _ (32)	6, 9, _, _ (40)
October 24	4, 6, _, _ (36)	_, _, 10, 14 (36)	3, _, 9, _ (32)
October 25	6, _, _, 12 (36)	_, 7, _, 15 (40)	3, _, _, 15 (36)
October 26	_, _, 12, 15 (40)	6, 9, _, _ (40)	_, _, 11, 14 (36)
Total	144	148	144

Step 2:
From condition (vi), we can conclude that the sales turnover of Barfi at Sector 70 on October 23, 24, 25 and 26 must be 9, 8, 7 and 6 (in Rs. '000s) respectively.
Now, we can see that the sales of each sweet at Sector 70 on October 25 were 3, 7, 11 and 15 (in Rs. '000s) which are all odd numbers. Except these values; all other values of the sales must be 2 even and 2 odd.

So the sales at Sector 35 on October 25 must be 6, 7, 12 and 15 (in Rs. '000s).
From condition (iii), we can conclude that the sales of Barfi at each outlet must be equal i.e., 7 (in Rs. '000s).
From condition (iv), we can conclude that the sales of Barfi at Sector 35 on each of the days October 23, 24 and 25 must be 7 (in Rs. '000s).
From condition (v), the sales of Rasmalai at Sector 70 on October 23 might be 12/14 (in Rs. '000s).
But if it is 12 (in Rs. '000s), then the sales of Rasgulla will be 13 (in Rs. '000s) which is not in the range.
So, the sales of sweets at Sector 70 on October 23 must be 6, 9, 11, 14 (in Rs. '000s).

Step 3:
From condition (vii), we can conclude that the sales of sweets at Sector 31 on October 24 would be 4, 6, 11 and 15 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 31 on October 23 would be 3, 6, 10 and 13 (in Rs. '000s).
It is very clear that the sales of sweets at Sector 35 on October 23 would be 4, 7, 9 and 12 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 35 on October 26 would be 6, 9, 11 and 14 (in Rs. '000s).
Hence, the final table can be shown as:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, 10, 13 (32)	4, 7, 9, 12 (32)	6, 9, 11, 14 (40)
October 24	4, 6, 11, 15 (36)	5, 7, 10, 14 (36)	3, 8, 9, 12 (32)
October 25	6, 7, 11, 12 (36)	6, 7, 12, 15 (40)	3, 7, 11, 15 (36)
October 26	4/5/6, 9/8/7, 12, 15 (40)	6, 9, 11, 14 (40)	5, 6, 11, 14 (36)
Total	144	148	144

The total sales of Rasmalai at Sector 35 on these 4 days was equal to 12 + 14 + 15 + 14 = 55 (in Rs. '000).

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Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Jaldiram has 3 sweet outlets in Sector 31, Sector 35 and Sector 70 of Gurgaon. He specializes in 4 types of sweets - Ladoos, Barfi, Rasgulla and Rasmalai. The following table gives partial information about the sales (in integral multiples of Rs. 1,000) of each outlet for the 4 consecutive days (from October 23 to October 26) during Diwali.

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _	4, _, _, _	6, 9, _, _
October 24	4, 6, _, _	_, _, 10, 14	3, _, 9, _
October 25	6, _, _, 12	_, 7, _, 15	3, _, _, 15
October 26	_, _, 12, 15	6, 9, _, _	_, _, 11, 14
Total	144	148	144

Note: In the above table, the sales figures are in the order - Ladoos, Barfi, Rasgulla, Rasmalai.
For example: The sales of Rasgulla and Rasmalai in Sector 35 outlet on October 24 were Rs. 10,000 and Rs. 14,000 respectively.
The sales of various sweets are in the range (in Rs. '000s) as per the table below:

Type of Sweets	Range
Ladoos	From 3 to 6
Barfi	From 6 to 9
Rasgulla	From 9 to 12
Rasmalai	From 12 to 15

For example: The sales of Ladoos in any outlet on any day can be equal to or more than Rs. 3,000 but less than or equal to Rs. 6,000.
The other known information are as below:
(i) The average of the sales of all the sweets is an integer (in integral multiples of Rs. 1,000), for each of the outlets on all the 4 days.
(ii) On each day exactly two of the outlets had the same total sales turnover. Each outlet had the same total turnover on exactly two out of the 4 days.
(iii) On October 25, each of the three outlets had the same sales turnover for exactly one of the sweets.
(iv) In Sector 35 outlet, On October 23rd, 24th and 25th the sales turnover of exactly one of the sweets was the same for all the three days.
(v) The total sales of Rasmalai on all the 4 days together, was an odd number for each of the three outlets.
(vi) The sales turnover of Barfi in Sector 70 on the 4 days was in an Arithmetic Progression.
(vii) The sales of the 4 types of sweets was not all even or all odd in any of the outlets on all the 4 days. There was only one exception to this rule on a particular day at one of the outlets.

Q.9 [11831809]

Which of the following statements are correct? (All in Rs. '000)

I. Total sales of Rasmalai on all the 4 days was the same for all the 3 outlets.

II. The total sales in all the 3 outlets together was the same on October 23 and October 24.

III. The total sales turnover in all the 3 outlets put together was maximum on October 25.

1 ◯ I & II only

2 ◯ II & III only

3 ◯ I & III only

4 ◯ All I, II & III

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
From the given range (in Rs. '000s) of the sales of various sweets, we can see that the maximum value of total sales at any outlet on any day = 6 + 9 + 12 + 15 = 42 and minimum value = 3 + 6 + 9 + 12 = 30.
From condition (i), we can conclude that the sales turnover (in Rs. '000s) at any outlet on any day may be 32/36/40.
From condition (iii), we can conclude the sales turnover (in Rs. '000s) as per below table range of the respective sales of sweets should also be kept in mind while preparing the below table:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _ (32)	4, _, _, _ (32)	6, 9, _, _ (40)
October 24	4, 6, _, _ (36)	_, _, 10, 14 (36)	3, _, 9, _ (32)
October 25	6, _, _, 12 (36)	_, 7, _, 15 (40)	3, _, _, 15 (36)
October 26	_, _, 12, 15 (40)	6, 9, _, _ (40)	_, _, 11, 14 (36)
Total	144	148	144

Step 2:
From condition (vi), we can conclude that the sales turnover of Barfi at Sector 70 on October 23, 24, 25 and 26 must be 9, 8, 7 and 6 (in Rs. '000s) respectively.
Now, we can see that the sales of each sweet at Sector 70 on October 25 were 3, 7, 11 and 15 (in Rs. '000s) which are all odd numbers. Except these values; all other values of the sales must be 2 even and 2 odd.

So the sales at Sector 35 on October 25 must be 6, 7, 12 and 15 (in Rs. '000s).
From condition (iii), we can conclude that the sales of Barfi at each outlet must be equal i.e., 7 (in Rs. '000s).
From condition (iv), we can conclude that the sales of Barfi at Sector 35 on each of the days October 23, 24 and 25 must be 7 (in Rs. '000s).
From condition (v), the sales of Rasmalai at Sector 70 on October 23 might be 12/14 (in Rs. '000s).
But if it is 12 (in Rs. '000s), then the sales of Rasgulla will be 13 (in Rs. '000s) which is not in the range.
So, the sales of sweets at Sector 70 on October 23 must be 6, 9, 11, 14 (in Rs. '000s).

Step 3:
From condition (vii), we can conclude that the sales of sweets at Sector 31 on October 24 would be 4, 6, 11 and 15 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 31 on October 23 would be 3, 6, 10 and 13 (in Rs. '000s).
It is very clear that the sales of sweets at Sector 35 on October 23 would be 4, 7, 9 and 12 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 35 on October 26 would be 6, 9, 11 and 14 (in Rs. '000s).
Hence, the final table can be shown as:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, 10, 13 (32)	4, 7, 9, 12 (32)	6, 9, 11, 14 (40)
October 24	4, 6, 11, 15 (36)	5, 7, 10, 14 (36)	3, 8, 9, 12 (32)
October 25	6, 7, 11, 12 (36)	6, 7, 12, 15 (40)	3, 7, 11, 15 (36)
October 26	4/5/6, 9/8/7, 12, 15 (40)	6, 9, 11, 14 (40)	5, 6, 11, 14 (36)
Total	144	148	144

Statements I & II are correct.

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Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Jaldiram has 3 sweet outlets in Sector 31, Sector 35 and Sector 70 of Gurgaon. He specializes in 4 types of sweets - Ladoos, Barfi, Rasgulla and Rasmalai. The following table gives partial information about the sales (in integral multiples of Rs. 1,000) of each outlet for the 4 consecutive days (from October 23 to October 26) during Diwali.

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _	4, _, _, _	6, 9, _, _
October 24	4, 6, _, _	_, _, 10, 14	3, _, 9, _
October 25	6, _, _, 12	_, 7, _, 15	3, _, _, 15
October 26	_, _, 12, 15	6, 9, _, _	_, _, 11, 14
Total	144	148	144

Note: In the above table, the sales figures are in the order - Ladoos, Barfi, Rasgulla, Rasmalai.
For example: The sales of Rasgulla and Rasmalai in Sector 35 outlet on October 24 were Rs. 10,000 and Rs. 14,000 respectively.
The sales of various sweets are in the range (in Rs. '000s) as per the table below:

Type of Sweets	Range
Ladoos	From 3 to 6
Barfi	From 6 to 9
Rasgulla	From 9 to 12
Rasmalai	From 12 to 15

For example: The sales of Ladoos in any outlet on any day can be equal to or more than Rs. 3,000 but less than or equal to Rs. 6,000.
The other known information are as below:
(i) The average of the sales of all the sweets is an integer (in integral multiples of Rs. 1,000), for each of the outlets on all the 4 days.
(ii) On each day exactly two of the outlets had the same total sales turnover. Each outlet had the same total turnover on exactly two out of the 4 days.
(iii) On October 25, each of the three outlets had the same sales turnover for exactly one of the sweets.
(iv) In Sector 35 outlet, On October 23rd, 24th and 25th the sales turnover of exactly one of the sweets was the same for all the three days.
(v) The total sales of Rasmalai on all the 4 days together, was an odd number for each of the three outlets.
(vi) The sales turnover of Barfi in Sector 70 on the 4 days was in an Arithmetic Progression.
(vii) The sales of the 4 types of sweets was not all even or all odd in any of the outlets on all the 4 days. There was only one exception to this rule on a particular day at one of the outlets.

Q.10 [11831809]

In which of the Outlet – Day combination was the sales of all the types of sweets either all odd numbers or all even numbers?

1 ◯ Sector 31 – October 25

2 ◯ Sector 31 – October 24

3 ◯ Sector 70 – October 24

4 ◯ Sector 70 – October 25

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
From the given range (in Rs. '000s) of the sales of various sweets, we can see that the maximum value of total sales at any outlet on any day = 6 + 9 + 12 + 15 = 42 and minimum value = 3 + 6 + 9 + 12 = 30.
From condition (i), we can conclude that the sales turnover (in Rs. '000s) at any outlet on any day may be 32/36/40.
From condition (iii), we can conclude the sales turnover (in Rs. '000s) as per below table range of the respective sales of sweets should also be kept in mind while preparing the below table:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, _, _ (32)	4, _, _, _ (32)	6, 9, _, _ (40)
October 24	4, 6, _, _ (36)	_, _, 10, 14 (36)	3, _, 9, _ (32)
October 25	6, _, _, 12 (36)	_, 7, _, 15 (40)	3, _, _, 15 (36)
October 26	_, _, 12, 15 (40)	6, 9, _, _ (40)	_, _, 11, 14 (36)
Total	144	148	144

Step 2:
From condition (vi), we can conclude that the sales turnover of Barfi at Sector 70 on October 23, 24, 25 and 26 must be 9, 8, 7 and 6 (in Rs. '000s) respectively.
Now, we can see that the sales of each sweet at Sector 70 on October 25 were 3, 7, 11 and 15 (in Rs. '000s) which are all odd numbers. Except these values; all other values of the sales must be 2 even and 2 odd.

So the sales at Sector 35 on October 25 must be 6, 7, 12 and 15 (in Rs. '000s).
From condition (iii), we can conclude that the sales of Barfi at each outlet must be equal i.e., 7 (in Rs. '000s).
From condition (iv), we can conclude that the sales of Barfi at Sector 35 on each of the days October 23, 24 and 25 must be 7 (in Rs. '000s).
From condition (v), the sales of Rasmalai at Sector 70 on October 23 might be 12/14 (in Rs. '000s).
But if it is 12 (in Rs. '000s), then the sales of Rasgulla will be 13 (in Rs. '000s) which is not in the range.
So, the sales of sweets at Sector 70 on October 23 must be 6, 9, 11, 14 (in Rs. '000s).

Step 3:
From condition (vii), we can conclude that the sales of sweets at Sector 31 on October 24 would be 4, 6, 11 and 15 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 31 on October 23 would be 3, 6, 10 and 13 (in Rs. '000s).
It is very clear that the sales of sweets at Sector 35 on October 23 would be 4, 7, 9 and 12 (in Rs. '000s).
From condition (v), the sales of sweets at Sector 35 on October 26 would be 6, 9, 11 and 14 (in Rs. '000s).
Hence, the final table can be shown as:

Outlet / Day	Sector 31	Sector 35	Sector 70
October 23	3, 6, 10, 13 (32)	4, 7, 9, 12 (32)	6, 9, 11, 14 (40)
October 24	4, 6, 11, 15 (36)	5, 7, 10, 14 (36)	3, 8, 9, 12 (32)
October 25	6, 7, 11, 12 (36)	6, 7, 12, 15 (40)	3, 7, 11, 15 (36)
October 26	4/5/6, 9/8/7, 12, 15 (40)	6, 9, 11, 14 (40)	5, 6, 11, 14 (36)
Total	144	148	144

The sales turnover at Sector 70 on October 25 was all odd numbers i.e., 3, 7, 11, 15 (in Rs. '000).

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Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

Mr. Mehta, an investor, had invested in 8 companies in 4 sectors – Consumer Services, Finance, Retail and Technology. The table given below shows the change in share prices of 8 companies disguised as A to H in no particular order. Mr. Mehta had bought shares in multiples of 10 on 1st November 2021 at the price mentioned in the table.

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022
A	160	220
B	400	475
C	200	320
D	350	420
E	420	560
F	250	300
G	180	240
H	480	540

Q.11 [11831809]

If Mr. Mehta had bought a total of 90 shares of 3 of the mentioned companies, then the maximum possible return on investment (in Rs.) as on 1st November 2022 based on the shares of those 3 companies could be

Solution:
Correct Answer : 11750

Answer key/Solution

The absolute and percentage variation of the shares are as follows:

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022	Absolute Variation (in Rs.)	Percentage Variation
A	160	220	60	37.50%
B	400	475	75	18.75%
C	200	320	120	60%
D	350	420	70	20%
E	420	560	140	33.33%
F	250	300	50	20%
G	180	240	60	33.33%
H	480	540	60	12.50%

Mr. Mehta had bought 90 shares across 3 companies and we need to calculate the maximum return. This is possible if he had bought 70 shares giving the maximum return i.e., of company E and minimum 10 shares each of the remaining 2 companies providing the next higher returns i.e., 10 shares each of companies C and B.
Hence, maximum return = (560 – 420) × 70 + (320 – 200) × 10 + (475 – 400) × 10 = Rs. 11,750.

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Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

Mr. Mehta, an investor, had invested in 8 companies in 4 sectors – Consumer Services, Finance, Retail and Technology. The table given below shows the change in share prices of 8 companies disguised as A to H in no particular order. Mr. Mehta had bought shares in multiples of 10 on 1st November 2021 at the price mentioned in the table.

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022
A	160	220
B	400	475
C	200	320
D	350	420
E	420	560
F	250	300
G	180	240
H	480	540

Q.12 [11831809]

If Mr. Mehta had invested Rs. 16,200 in companies whose share prices appreciated by the same amount, then what could be the maximum number of shares of these companies he bought?

Solution:

Correct Answer : 80

Answer key/Solution

The absolute and percentage variation of the shares are as follows:

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022	Absolute Variation (in Rs.)	Percentage Variation
A	160	220	60	37.50%
B	400	475	75	18.75%
C	200	320	120	60%
D	350	420	70	20%
E	420	560	140	33.33%
F	250	300	50	20%
G	180	240	60	33.33%
H	480	540	60	12.50%

Companies A, G and H showed the equal absolute variation.

(160 + 180 + 480) × 10 + 160 × 50 = Rs. 16,200

So for maximum shares, he had bought 60, 10 and 10 shares of company A, G and H respectively.

Hence, required maximum number of shares bought by him = 60 + 10 + 10 = 80.

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Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

Mr. Mehta, an investor, had invested in 8 companies in 4 sectors – Consumer Services, Finance, Retail and Technology. The table given below shows the change in share prices of 8 companies disguised as A to H in no particular order. Mr. Mehta had bought shares in multiples of 10 on 1st November 2021 at the price mentioned in the table.

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022
A	160	220
B	400	475
C	200	320
D	350	420
E	420	560
F	250	300
G	180	240
H	480	540

Q.13 [11831809]

If one of the two Retail companies showed the highest percentage variation in the share price while the other Retail company showed the lowest percentage variation, then the two Retail companies were

1 ◯ C & F

2 ◯ A & D

3 ◯ C & H

4 ◯ E & H

Solution:
Correct Answer : 3

Answer key/Solution

The absolute and percentage variation of the shares are as follows:

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022	Absolute Variation (in Rs.)	Percentage Variation
A	160	220	60	37.50%
B	400	475	75	18.75%
C	200	320	120	60%
D	350	420	70	20%
E	420	560	140	33.33%
F	250	300	50	20%
G	180	240	60	33.33%
H	480	540	60	12.50%

Company C showed the highest percentage variation i.e., (320 – 200)/200 × 100 = 60%
Company H showed the lowest percentage variation i.e., (540 – 480)/480 × 100 = 12.5%
Hence, the two Retail companies were C and H.

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Directions for questions 11 to 14: Answer the questions on the basis of the information given below.

Mr. Mehta, an investor, had invested in 8 companies in 4 sectors – Consumer Services, Finance, Retail and Technology. The table given below shows the change in share prices of 8 companies disguised as A to H in no particular order. Mr. Mehta had bought shares in multiples of 10 on 1st November 2021 at the price mentioned in the table.

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022
A	160	220
B	400	475
C	200	320
D	350	420
E	420	560
F	250	300
G	180	240
H	480	540

Q.14 [11831809]

Mr. Mehta had invested money in the ratio 3 : 2 in companies C and D while the number of shares of C and D bought by his friend Mr. Gupta was in the ratio 3 : 2. Both these purchases were made on November 1, 2021. If they both of them did not sell any part of this investment, then which of the following statements is true as on November 1, 2022?

1 ◯ The return on this investment was the same for both.

2 ◯ The return on investment for Mr. Gupta was about 39%.

3 ◯ The return on investment for Mr. Mehta was 36%.

4 ◯ The return on investment for Mr. Gupta was 44%.

Solution:
Correct Answer : 2

Answer key/Solution

The absolute and percentage variation of the shares are as follows:

Company	Share Prices (in Rs.) as on 1st November 2021	Share Prices (in Rs.) as on 1st November 2022	Absolute Variation (in Rs.)	Percentage Variation
A	160	220	60	37.50%
B	400	475	75	18.75%
C	200	320	120	60%
D	350	420	70	20%
E	420	560	140	33.33%
F	250	300	50	20%
G	180	240	60	33.33%
H	480	540	60	12.50%

Let Mr. Mehta had invested Rs. 300 and Rs. 200 in companies C and D respectively.
Then, his total return on November 1, 2022 = 300 × 1.6 + 200 × 1.2 = Rs. 720
So his percentage return on investment = (720 - 500)/500 = 44%
Let the number of shares of C and D bought by Mr. Gupta be 3 and 2 respectively.
Then, his total investment = 3 × 200 + 2 × 350 = Rs.1,300
Then, his total return on November 1, 2022 = 600 × 1.6 + 700 × 1.2 = Rs. 1,800
So his percentage return on investment = (1800 – 1300)/1300 = 38.46% ≈ 39%.
Hence, the statement given in option (2) is true.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

The ZED-Detective Group has seven secret agents - DHRUV, JAZEL, PARAG, SALIM, TAPAK, VIREN and ZAMIR. The organization does not use their names under any circumstances.
The names are coded and their secret duties are assigned at seven different airports - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bangalore and Lucknow - not necessarily in this order. The coding for the names is based on a unique code that is assigned to every alphabet in English. The consonants are coded as their respective position in the order of the English alphabets, while each of the vowels is given a code that represents its position in the alphabetical order of the vowels. Once their codes are assigned, the agents are addressed by their codes and never by their names. In the organization, there are two sniffer dogs, namely RAVLO and ZIBRA and it is known that the codes for their names are 18-1-22-12-4 and 26-3-2-18-1 respectively. The rules for coding the agents' names are the same as those for the dogs' names. Each of these agents intercepted exactly one illegal item among - Gold, Weapons, Explosives and Drugs.
Further, it is also known that:
(i) The code with the smallest sum of digits is assigned duty at Delhi airport, where Weapons were intercepted.
(ii) The agent having a code in which the digits add up to the same number as the sum of digits in a dog's code takes the dog along. It is known that the dog with an even number of digits in its code goes to Mumbai whereas the other dog does not go to Lucknow.
(iii) The agent with a code that is an even number but has an odd number of digits is assigned duty at Hyderabad. The agent with a code that has some of the digits in the given order **1133** is assigned Bangalore.
(iv) The agent with a code having an odd number of digits with the sum of digits divisible by 5 is assigned duty at Kolkata, where Drugs were intercepted. The other agent who intercepted Drugs had a code starting with 48 and ending with 22.
(v) Gold was intercepted by agents having codes with last two digits as 14 and 18 whereas Explosives were intercepted by an agent with a sniffer dog and also, by the agent at Lucknow.

Q.15 [11831809]
How many agents have codes that are divisible by 3?

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
From the given information and condition (ii), On observing the codes for the names of dogs we can see that the rule followed is quite simple.
The consonants are coded with their respective positions in the English alphabet.
The vowels - A, E, I, O and U - are assigned numbers 1, 2, 3, 4 and 5 respectively.
So the names are coded as follows:

R	A	V	L	O
18	1	22	12	4

There are an even number of digits, the code is an odd number and the sum of digits is 21. RAVLO goes to Mumbai.

Z	I	B	R	A
26	3	2	18	1

There are an odd number of digits, the code is an odd number and the sum of digits is 23. ZIBRA goes with the agent whose code has the sum of digits equal to 23.

Step 2:
From conditions (i), (iii), (iv) and (v), Let us write the codes for the names of the agents and find out in which city they are assigned duties.
Hence, the final table can be shown as:

						Sum of digits	Code (Even/Odd) No. of digits	Place	Item intercepted
Name	D	H	R	U	V	30	Even - 7 digits	Hyderabad	Drugs
Code	4	8	18	5	22				
Name	Z	A	M	I	R	25	Even - 8 digits	Bangalore	Gold
Code	26	1	13	3	18				
Name	T	A	P	A	K	13	Odd - 8 digits	Delhi	Weapons
Code	20	1	16	1	11				
Name	S	A	L	I	M	21	Odd - 8 digits	Mumbai (RAVLO)	Explosives
Code	19	1	12	3	13				
Name	V	I	R	E	N	23	Even - 8 digits	Chennai (ZIBRA)	Gold
Code	22	3	18	2	14				
Name	P	A	R	A	G	25	Odd - 7 digits	Kolkata	Drugs
Code	16	1	18	1	7				
Name	J	A	Z	E	L	15	Even - 8 digits	Lucknow	Explosives
Code	10	1	26	2	12				

Any number is divisible by 3 if the sum of digits is divisible by 3. We can see from the table that the sum of digits of three codes are 30, 21 and 15 respectively.. Hence, three out of seven codes are divisible by 3.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

The ZED-Detective Group has seven secret agents - DHRUV, JAZEL, PARAG, SALIM, TAPAK, VIREN and ZAMIR. The organization does not use their names under any circumstances.
The names are coded and their secret duties are assigned at seven different airports - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bangalore and Lucknow - not necessarily in this order. The coding for the names is based on a unique code that is assigned to every alphabet in English. The consonants are coded as their respective position in the order of the English alphabets, while each of the vowels is given a code that represents its position in the alphabetical order of the vowels. Once their codes are assigned, the agents are addressed by their codes and never by their names. In the organization, there are two sniffer dogs, namely RAVLO and ZIBRA and it is known that the codes for their names are 18-1-22-12-4 and 26-3-2-18-1 respectively. The rules for coding the agents' names are the same as those for the dogs' names. Each of these agents intercepted exactly one illegal item among - Gold, Weapons, Explosives and Drugs.
Further, it is also known that:
(i) The code with the smallest sum of digits is assigned duty at Delhi airport, where Weapons were intercepted.
(ii) The agent having a code in which the digits add up to the same number as the sum of digits in a dog's code takes the dog along. It is known that the dog with an even number of digits in its code goes to Mumbai whereas the other dog does not go to Lucknow.
(iii) The agent with a code that is an even number but has an odd number of digits is assigned duty at Hyderabad. The agent with a code that has some of the digits in the given order **1133** is assigned Bangalore.
(iv) The agent with a code having an odd number of digits with the sum of digits divisible by 5 is assigned duty at Kolkata, where Drugs were intercepted. The other agent who intercepted Drugs had a code starting with 48 and ending with 22.
(v) Gold was intercepted by agents having codes with last two digits as 14 and 18 whereas Explosives were intercepted by an agent with a sniffer dog and also, by the agent at Lucknow.

Q.16 [11831809]
What are the codes of the agents who are assigned duties at Kolkata and Lucknow respectively?

1 ◯ 19112313 - 26113318

2 ◯ 20116111 - 10126212

3 ◯ 1611817 - 10126212

4 ◯ 1611817 - 22318214

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
From the given information and condition (ii), On observing the codes for the names of dogs we can see that the rule followed is quite simple.
The consonants are coded with their respective positions in the English alphabet.
The vowels - A, E, I, O and U - are assigned numbers 1, 2, 3, 4 and 5 respectively.
So the names are coded as follows:

R	A	V	L	O
18	1	22	12	4

There are an even number of digits, the code is an odd number and the sum of digits is 21. RAVLO goes to Mumbai.

Z	I	B	R	A
26	3	2	18	1

There are an odd number of digits, the code is an odd number and the sum of digits is 23. ZIBRA goes with the agent whose code has the sum of digits equal to 23.

Step 2:
From conditions (i), (iii), (iv) and (v), Let us write the codes for the names of the agents and find out in which city they are assigned duties.
Hence, the final table can be shown as:

						Sum of digits	Code (Even/Odd) No. of digits	Place	Item intercepted
Name	D	H	R	U	V	30	Even - 7 digits	Hyderabad	Drugs
Code	4	8	18	5	22				
Name	Z	A	M	I	R	25	Even - 8 digits	Bangalore	Gold
Code	26	1	13	3	18				
Name	T	A	P	A	K	13	Odd - 8 digits	Delhi	Weapons
Code	20	1	16	1	11				
Name	S	A	L	I	M	21	Odd - 8 digits	Mumbai (RAVLO)	Explosives
Code	19	1	12	3	13				
Name	V	I	R	E	N	23	Even - 8 digits	Chennai (ZIBRA)	Gold
Code	22	3	18	2	14				
Name	P	A	R	A	G	25	Odd - 7 digits	Kolkata	Drugs
Code	16	1	18	1	7				
Name	J	A	Z	E	L	15	Even - 8 digits	Lucknow	Explosives
Code	10	1	26	2	12				

We can see that PARAG and JAZEL with codes 1611817 and 10126212 go to Kolkata and Lucknow respectively.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

The ZED-Detective Group has seven secret agents - DHRUV, JAZEL, PARAG, SALIM, TAPAK, VIREN and ZAMIR. The organization does not use their names under any circumstances.
The names are coded and their secret duties are assigned at seven different airports - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bangalore and Lucknow - not necessarily in this order. The coding for the names is based on a unique code that is assigned to every alphabet in English. The consonants are coded as their respective position in the order of the English alphabets, while each of the vowels is given a code that represents its position in the alphabetical order of the vowels. Once their codes are assigned, the agents are addressed by their codes and never by their names. In the organization, there are two sniffer dogs, namely RAVLO and ZIBRA and it is known that the codes for their names are 18-1-22-12-4 and 26-3-2-18-1 respectively. The rules for coding the agents' names are the same as those for the dogs' names. Each of these agents intercepted exactly one illegal item among - Gold, Weapons, Explosives and Drugs.
Further, it is also known that:
(i) The code with the smallest sum of digits is assigned duty at Delhi airport, where Weapons were intercepted.
(ii) The agent having a code in which the digits add up to the same number as the sum of digits in a dog's code takes the dog along. It is known that the dog with an even number of digits in its code goes to Mumbai whereas the other dog does not go to Lucknow.
(iii) The agent with a code that is an even number but has an odd number of digits is assigned duty at Hyderabad. The agent with a code that has some of the digits in the given order **1133** is assigned Bangalore.
(iv) The agent with a code having an odd number of digits with the sum of digits divisible by 5 is assigned duty at Kolkata, where Drugs were intercepted. The other agent who intercepted Drugs had a code starting with 48 and ending with 22.
(v) Gold was intercepted by agents having codes with last two digits as 14 and 18 whereas Explosives were intercepted by an agent with a sniffer dog and also, by the agent at Lucknow.

Q.17 [11831809]
What is the name of the agent with whom ZIBRA goes on duty?

1 ◯ PARAG

2 ◯ VIREN

3 ◯ JAZEL

4 ◯ SALIM

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
From the given information and condition (ii), On observing the codes for the names of dogs we can see that the rule followed is quite simple.
The consonants are coded with their respective positions in the English alphabet.
The vowels - A, E, I, O and U - are assigned numbers 1, 2, 3, 4 and 5 respectively.
So the names are coded as follows:

R	A	V	L	O
18	1	22	12	4

There are an even number of digits, the code is an odd number and the sum of digits is 21. RAVLO goes to Mumbai.

Z	I	B	R	A
26	3	2	18	1

There are an odd number of digits, the code is an odd number and the sum of digits is 23. ZIBRA goes with the agent whose code has the sum of digits equal to 23.

Step 2:
From conditions (i), (iii), (iv) and (v), Let us write the codes for the names of the agents and find out in which city they are assigned duties.
Hence, the final table can be shown as:

						Sum of digits	Code (Even/Odd) No. of digits	Place	Item intercepted
Name	D	H	R	U	V	30	Even - 7 digits	Hyderabad	Drugs
Code	4	8	18	5	22				
Name	Z	A	M	I	R	25	Even - 8 digits	Bangalore	Gold
Code	26	1	13	3	18				
Name	T	A	P	A	K	13	Odd - 8 digits	Delhi	Weapons
Code	20	1	16	1	11				
Name	S	A	L	I	M	21	Odd - 8 digits	Mumbai (RAVLO)	Explosives
Code	19	1	12	3	13				
Name	V	I	R	E	N	23	Even - 8 digits	Chennai (ZIBRA)	Gold
Code	22	3	18	2	14				
Name	P	A	R	A	G	25	Odd - 7 digits	Kolkata	Drugs
Code	16	1	18	1	7				
Name	J	A	Z	E	L	15	Even - 8 digits	Lucknow	Explosives
Code	10	1	26	2	12				

The sum of digits of ZIBRA's code is 23 and we can see from the table that the sum of digits of VIREN's code is also 23. Hence, ZIBRA goes to Chennai with VIREN on duty.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

The ZED-Detective Group has seven secret agents - DHRUV, JAZEL, PARAG, SALIM, TAPAK, VIREN and ZAMIR. The organization does not use their names under any circumstances.
The names are coded and their secret duties are assigned at seven different airports - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bangalore and Lucknow - not necessarily in this order. The coding for the names is based on a unique code that is assigned to every alphabet in English. The consonants are coded as their respective position in the order of the English alphabets, while each of the vowels is given a code that represents its position in the alphabetical order of the vowels. Once their codes are assigned, the agents are addressed by their codes and never by their names. In the organization, there are two sniffer dogs, namely RAVLO and ZIBRA and it is known that the codes for their names are 18-1-22-12-4 and 26-3-2-18-1 respectively. The rules for coding the agents' names are the same as those for the dogs' names. Each of these agents intercepted exactly one illegal item among - Gold, Weapons, Explosives and Drugs.
Further, it is also known that:
(i) The code with the smallest sum of digits is assigned duty at Delhi airport, where Weapons were intercepted.
(ii) The agent having a code in which the digits add up to the same number as the sum of digits in a dog's code takes the dog along. It is known that the dog with an even number of digits in its code goes to Mumbai whereas the other dog does not go to Lucknow.
(iii) The agent with a code that is an even number but has an odd number of digits is assigned duty at Hyderabad. The agent with a code that has some of the digits in the given order **1133** is assigned Bangalore.
(iv) The agent with a code having an odd number of digits with the sum of digits divisible by 5 is assigned duty at Kolkata, where Drugs were intercepted. The other agent who intercepted Drugs had a code starting with 48 and ending with 22.
(v) Gold was intercepted by agents having codes with last two digits as 14 and 18 whereas Explosives were intercepted by an agent with a sniffer dog and also, by the agent at Lucknow.

Q.18 [11831809]
Duty at which airport is assigned to the agent having a code ending in 18?

1 ◯ Kolkata

2 ◯ Lucknow

3 ◯ Chennai

4 ◯ Bangalore

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
From the given information and condition (ii), On observing the codes for the names of dogs we can see that the rule followed is quite simple.
The consonants are coded with their respective positions in the English alphabet.
The vowels - A, E, I, O and U - are assigned numbers 1, 2, 3, 4 and 5 respectively.
So the names are coded as follows:

R	A	V	L	O
18	1	22	12	4

There are an even number of digits, the code is an odd number and the sum of digits is 21. RAVLO goes to Mumbai.

Z	I	B	R	A
26	3	2	18	1

There are an odd number of digits, the code is an odd number and the sum of digits is 23. ZIBRA goes with the agent whose code has the sum of digits equal to 23.

Step 2:
From conditions (i), (iii), (iv) and (v), Let us write the codes for the names of the agents and find out in which city they are assigned duties.
Hence, the final table can be shown as:

						Sum of digits	Code (Even/Odd) No. of digits	Place	Item intercepted
Name	D	H	R	U	V	30	Even - 7 digits	Hyderabad	Drugs
Code	4	8	18	5	22				
Name	Z	A	M	I	R	25	Even - 8 digits	Bangalore	Gold
Code	26	1	13	3	18				
Name	T	A	P	A	K	13	Odd - 8 digits	Delhi	Weapons
Code	20	1	16	1	11				
Name	S	A	L	I	M	21	Odd - 8 digits	Mumbai (RAVLO)	Explosives
Code	19	1	12	3	13				
Name	V	I	R	E	N	23	Even - 8 digits	Chennai (ZIBRA)	Gold
Code	22	3	18	2	14				
Name	P	A	R	A	G	25	Odd - 7 digits	Kolkata	Drugs
Code	16	1	18	1	7				
Name	J	A	Z	E	L	15	Even - 8 digits	Lucknow	Explosives
Code	10	1	26	2	12				

We can see from the table that 26113318 is the code given to ZAMIR, who is assigned duty at Bangalore.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

The ZED-Detective Group has seven secret agents - DHRUV, JAZEL, PARAG, SALIM, TAPAK, VIREN and ZAMIR. The organization does not use their names under any circumstances.
The names are coded and their secret duties are assigned at seven different airports - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bangalore and Lucknow - not necessarily in this order. The coding for the names is based on a unique code that is assigned to every alphabet in English. The consonants are coded as their respective position in the order of the English alphabets, while each of the vowels is given a code that represents its position in the alphabetical order of the vowels. Once their codes are assigned, the agents are addressed by their codes and never by their names. In the organization, there are two sniffer dogs, namely RAVLO and ZIBRA and it is known that the codes for their names are 18-1-22-12-4 and 26-3-2-18-1 respectively. The rules for coding the agents' names are the same as those for the dogs' names. Each of these agents intercepted exactly one illegal item among - Gold, Weapons, Explosives and Drugs.
Further, it is also known that:
(i) The code with the smallest sum of digits is assigned duty at Delhi airport, where Weapons were intercepted.
(ii) The agent having a code in which the digits add up to the same number as the sum of digits in a dog's code takes the dog along. It is known that the dog with an even number of digits in its code goes to Mumbai whereas the other dog does not go to Lucknow.
(iii) The agent with a code that is an even number but has an odd number of digits is assigned duty at Hyderabad. The agent with a code that has some of the digits in the given order **1133** is assigned Bangalore.
(iv) The agent with a code having an odd number of digits with the sum of digits divisible by 5 is assigned duty at Kolkata, where Drugs were intercepted. The other agent who intercepted Drugs had a code starting with 48 and ending with 22.
(v) Gold was intercepted by agents having codes with last two digits as 14 and 18 whereas Explosives were intercepted by an agent with a sniffer dog and also, by the agent at Lucknow.

Q.19 [11831809]
What was the item intercepted by the team of which RAVLO is a part?

1 ◯ Explosives

2 ◯ Weapons

3 ◯ Gold

4 ◯ Drugs

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
From the given information and condition (ii), On observing the codes for the names of dogs we can see that the rule followed is quite simple.
The consonants are coded with their respective positions in the English alphabet.
The vowels - A, E, I, O and U - are assigned numbers 1, 2, 3, 4 and 5 respectively.
So the names are coded as follows:

R	A	V	L	O
18	1	22	12	4

There are an even number of digits, the code is an odd number and the sum of digits is 21. RAVLO goes to Mumbai.

Z	I	B	R	A
26	3	2	18	1

There are an odd number of digits, the code is an odd number and the sum of digits is 23. ZIBRA goes with the agent whose code has the sum of digits equal to 23.

Step 2:
From conditions (i), (iii), (iv) and (v), Let us write the codes for the names of the agents and find out in which city they are assigned duties.
Hence, the final table can be shown as:

						Sum of digits	Code (Even/Odd) No. of digits	Place	Item intercepted
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Code	26	1	13	3	18				
Name	T	A	P	A	K	13	Odd - 8 digits	Delhi	Weapons
Code	20	1	16	1	11				
Name	S	A	L	I	M	21	Odd - 8 digits	Mumbai (RAVLO)	Explosives
Code	19	1	12	3	13				
Name	V	I	R	E	N	23	Even - 8 digits	Chennai (ZIBRA)	Gold
Code	22	3	18	2	14				
Name	P	A	R	A	G	25	Odd - 7 digits	Kolkata	Drugs
Code	16	1	18	1	7				
Name	J	A	Z	E	L	15	Even - 8 digits	Lucknow	Explosives
Code	10	1	26	2	12				

RAVLO is a part of agent SALIM's team, which is assigned duty at Mumbai airport, where Explosives were intercepted.

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Directions for questions 15 to 20: Answer the questions on the basis of the information given below.

The ZED-Detective Group has seven secret agents - DHRUV, JAZEL, PARAG, SALIM, TAPAK, VIREN and ZAMIR. The organization does not use their names under any circumstances.
The names are coded and their secret duties are assigned at seven different airports - Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Bangalore and Lucknow - not necessarily in this order. The coding for the names is based on a unique code that is assigned to every alphabet in English. The consonants are coded as their respective position in the order of the English alphabets, while each of the vowels is given a code that represents its position in the alphabetical order of the vowels. Once their codes are assigned, the agents are addressed by their codes and never by their names. In the organization, there are two sniffer dogs, namely RAVLO and ZIBRA and it is known that the codes for their names are 18-1-22-12-4 and 26-3-2-18-1 respectively. The rules for coding the agents' names are the same as those for the dogs' names. Each of these agents intercepted exactly one illegal item among - Gold, Weapons, Explosives and Drugs.
Further, it is also known that:
(i) The code with the smallest sum of digits is assigned duty at Delhi airport, where Weapons were intercepted.
(ii) The agent having a code in which the digits add up to the same number as the sum of digits in a dog's code takes the dog along. It is known that the dog with an even number of digits in its code goes to Mumbai whereas the other dog does not go to Lucknow.
(iii) The agent with a code that is an even number but has an odd number of digits is assigned duty at Hyderabad. The agent with a code that has some of the digits in the given order **1133** is assigned Bangalore.
(iv) The agent with a code having an odd number of digits with the sum of digits divisible by 5 is assigned duty at Kolkata, where Drugs were intercepted. The other agent who intercepted Drugs had a code starting with 48 and ending with 22.
(v) Gold was intercepted by agents having codes with last two digits as 14 and 18 whereas Explosives were intercepted by an agent with a sniffer dog and also, by the agent at Lucknow.

Q.20 [11831809]

Which of these agents intercepted Drugs at their location?
I. DHRUV
II. PARAG
III. TAPAK

1 ◯ I only

2 ◯ II & III only

3 ◯ All I, II & III

4 ◯ I & II only

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
From the given information and condition (ii), On observing the codes for the names of dogs we can see that the rule followed is quite simple.
The consonants are coded with their respective positions in the English alphabet.
The vowels - A, E, I, O and U - are assigned numbers 1, 2, 3, 4 and 5 respectively.
So the names are coded as follows:

R	A	V	L	O
18	1	22	12	4

There are an even number of digits, the code is an odd number and the sum of digits is 21. RAVLO goes to Mumbai.

Z	I	B	R	A
26	3	2	18	1

There are an odd number of digits, the code is an odd number and the sum of digits is 23. ZIBRA goes with the agent whose code has the sum of digits equal to 23.

Step 2:
From conditions (i), (iii), (iv) and (v), Let us write the codes for the names of the agents and find out in which city they are assigned duties.
Hence, the final table can be shown as:

						Sum of digits	Code (Even/Odd) No. of digits	Place	Item intercepted
Name	D	H	R	U	V	30	Even - 7 digits	Hyderabad	Drugs
Code	4	8	18	5	22				
Name	Z	A	M	I	R	25	Even - 8 digits	Bangalore	Gold
Code	26	1	13	3	18				
Name	T	A	P	A	K	13	Odd - 8 digits	Delhi	Weapons
Code	20	1	16	1	11				
Name	S	A	L	I	M	21	Odd - 8 digits	Mumbai (RAVLO)	Explosives
Code	19	1	12	3	13				
Name	V	I	R	E	N	23	Even - 8 digits	Chennai (ZIBRA)	Gold
Code	22	3	18	2	14				
Name	P	A	R	A	G	25	Odd - 7 digits	Kolkata	Drugs
Code	16	1	18	1	7				
Name	J	A	Z	E	L	15	Even - 8 digits	Lucknow	Explosives
Code	10	1	26	2	12				

We can observe from the table that both DHRUV and PARAG intercepted Drugs at their locations.

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